Rock-Paper-Scissors Game

Week 1 Practice - Problem 1 | Java

Problem Summary

Simulate five rounds. The computer chooses randomly, the player’s example moves are predefined for a live demo, and the program reports the per-round table and summary statistics.

Java Solution

```
import java.util.Random;
public class Main {
    static String playRound(String playerMove, String computerMove) {
        if (playerMove.equals(computerMove)) return "Draw";
        if ((playerMove.equals("Rock") && computerMove.equals("Scissors"))
            || (playerMove.equals("Paper") && computerMove.equals("Rock"))
            || (playerMove.equals("Scissors") && computerMove.equals("Paper")))
            return "Player Wins";
        return "Computer Wins";
    }
    public static void main(String[] args) {
        String[] moves = {"Rock", "Paper", "Scissors"};
        String[] player = {"Rock", "Paper", "Scissors", "Rock", "Paper"};
        Random random = new Random();
        int wins = 0, losses = 0, draws = 0;
        System.out.println("Round | Player | Computer | Result");
        for (int i = 0; i < player.length; i++) {
            String computer = moves[random.nextInt(3)];
            String result = playRound(player[i], computer);
            if (result.equals("Player Wins")) wins++;
            else if (result.equals("Computer Wins")) losses++;
            else draws++;
            System.out.printf("%d | %s | %s | %s%n",
                i + 1, player[i], computer, result);
        }
        System.out.printf("Wins: %d | Losses: %d | Draws: %d | Win %% = %.1f%%%n",
            wins, losses, draws, wins * 100.0 / player.length);
    }
}
```

Sample Output

```
Round | Player | Computer | Result
(Computer moves are random; sample rows vary.)
Wins + Losses + Draws = 5
Win percentage = Wins / 5 * 100
```

Complexity / Key Notes

Time and space: O(N), where N is the number of rounds, because the moves/results are processed and kept for the report. Player input is represented by a predefined list as allowed in the prompt.